

Day 20 (02/12/2025)

Dictionary - It is also multivalued heterogeneous data type store data in key-value pairs. and it is unordered, mutable, duplicates allowed only in values, but duplicate keys are not allowed.
And these are accessing using keys, not by indexes.

key-value pairs means -

key $\rightarrow$ name/label

value $\rightarrow$ data stored for that key

ex- just like how we save a contact in our phone:

Name : "poojtha"

Number : 1234567890

it is represented by {key: value}

dl = { "name" : "poojyy", "email" : "poojtha@gmail.com", "age" : 23,
dl "place" : "Bengaluru" }

~~dl~~ { 'name' : 'poojyy',
'email' : 'poojtha@gmail.com',
'age' : 23,
'place' : 'Bengaluru' }

dl["name"]

o/p $\rightarrow$ 'poojyy'

→ `dl.keys()` ⇒ for all keys

o/p `dict_keys(['name', 'email', 'age', 'place'])`

→ `dl.values()` → for all values

o/p `dict_values(['poojyy', 'poojtha@gmail.com', 23, 'Bengaluru'])`

→ `dl.items()` → for all key-value pairs -

o/p `dict_items([('name', 'poojyy'), ('email', 'poojtha@gmail.com'), ('age', 23), ('place', 'Bengaluru')])`

→ adding element -

`dl["course"] = "Data Analyst"`

o/p `dl`
`{ 'name' : 'poojyy',`
 `'email' : 'poojtha@gmail.com',`
 `'age' : 23,`
 `'place' : 'Bengaluru',`
 `'course' : 'Data Analyst' }`

→ updating element -

`dl["age"] = 28`

o/p `dl`
`{ 'name' : 'poojyy',`
 `'email' : 'poojtha@gmail.com',`
 `'age' : 28,`
 `'place' : 'Bengaluru',`
 `'course' : 'Data Analyst' }`

→ Deleting element -

`dl.pop("age")`

`dl`

o/p
{ 'name' : 'poorjya',
 'email' : 'poorjtha@gmail.com',
 'place' : 'Bengaluru',
 'Course' : 'Data Analyst' }

→ `dl.popitem()` → remove last key-value.

`dl`

o/p { 'name' : 'poorjya',
 'email' : 'poorjtha@gmail.com',
 'place' : 'Bengaluru' } ⇒ as ~~key~~ course is deleted.

Q) Value = "PriYaNKA123@7&0*!"

Count upper char, lower char, numbers, spec char.

Value = "PriYaNKA123@7&0*!"

`uc = 0`

`lc = 0`

`nc = 0`

`sc = 0`

for i in value:

if i.isupper():

`uc = uc + 1`

elif i.islower():

`lc = lc + 1`

elif i.isdigit():

`nc = nc + 1`

else:

`sc = sc + 1`

`print(f"upper = {uc} | lower = {lc} | number = {nc} | special = {sc}")`

upper = 5

lower = 3

number = 5

special = 4